ROCSCIENCE

Case8
Slide2 - An Interactive Slope Stability Program
Date Created: 27/03/2026, 9:23:52 pm
Software Version: 9.041

Table of Contents

Project Summary 3
General Settings 4
Analysis Options 5
Groundwater Analysis 6
Random Numbers 7
Surface Options 8
Seismic Loading 9
Materials 10
Global Minimums 11
- Method: bishop simplified 11
- Method: spencer 11

Valid and Invalid Surfaces 12
- Method: bishop simplified 12
- Method: spencer 12
  - Error Code Descriptions 12

Slice Data 13
- Global Minimum Query (bishop simplified) - Safety Factor: 2.51162 13
- Query 1 (bishop simplified) - Safety Factor: 2.51162 14
- Global Minimum Query (spencer) - Safety Factor: 2.50591 15
- Query 1 (spencer) - Safety Factor: 2.50591 16

Interslice Data 17
- Global Minimum Query (bishop simplified) - Safety Factor: 2.51162 17
- Query 1 (bishop simplified) - Safety Factor: 2.51162 18
- Global Minimum Query (spencer) - Safety Factor: 2.50591 19
- Query 1 (spencer) - Safety Factor: 2.50591 20

Entity Information 21
- Group 1 21
  - Shared Entities 21
  - Scenario-based Entities 21

Slide2 Analysis Information

Case8

Project Summary

File Name:	Case8.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.718s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	27/03/2026, 9:23:52 pm

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.001
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

TBA	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	12 kPa
Phi	38 °
Water Surface	Water Table
Hu Type	Automatically Calculated
Specify alternate strength type above water surface	No

Global Minimums

Method: bishop simplified

FS	2.511620
Center:	5.812, 13.042
Radius:	10.075
Left Slip Surface Endpoint:	5.000, 3.000
Right Slip Surface Endpoint:	15.040, 9.000
Left Slope Intercept:	5.000 7.184
Right Slope Intercept:	15.040 9.000
Resisting Moment:	2871.19 kN-m
Driving Moment:	1143.16 kN-m
Total Slice Area:	34.2074 m2
Surface Horizontal Width:	10.0399 m
Surface Average Height:	3.40714 m

Method: spencer

FS	2.505910
Center:	5.812, 13.042
Radius:	10.075
Left Slip Surface Endpoint:	5.000, 3.000
Right Slip Surface Endpoint:	15.040, 9.000
Left Slope Intercept:	5.000 7.184
Right Slope Intercept:	15.040 9.000
Resisting Moment:	2864.66 kN-m
Driving Moment:	1143.16 kN-m
Resisting Horizontal Force:	234.436 kN
Driving Horizontal Force:	93.553 kN
Total Slice Area:	34.2074 m2
Surface Horizontal Width:	10.0399 m
Surface Average Height:	3.40714 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	8376
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	8194
Number of Invalid Surfaces:	182

Error Codes

Error Code -108 reported for 115 surfaces
Error Code -111 reported for 61 surfaces
Error Code -112 reported for 6 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.51162

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.198879	8.2906	-4.05273	TBA	12	38	5.06716	12.7268	42.0457	41.1155	0.930247	41.6867	0.57123
2	0.198879	8.53903	-2.91954	TBA	12	38	5.3926	13.5442	43.2109	41.2344	1.9765	42.9359	1.70148
3	0.198879	8.77491	-1.7875	TBA	12	38	5.70643	14.3324	44.3001	41.3147	2.98535	44.122	2.80727
4	0.198879	8.99829	-0.656162	TBA	12	38	6.00884	15.0919	45.3139	41.3564	3.95753	45.2451	3.88871
5	0.198879	9.20916	0.474923	TBA	12	38	6.30004	15.8233	46.2532	41.3596	4.89362	46.3054	4.94584
6	0.198879	9.40754	1.60619	TBA	12	38	6.58017	16.5269	47.1184	41.3243	5.79413	47.3029	5.97865
7	0.198879	9.59341	2.73809	TBA	12	38	6.84938	17.203	47.91	41.2504	6.65959	48.2376	6.98716
8	0.198879	9.76674	3.87106	TBA	12	38	7.10779	17.8521	48.6281	41.1378	7.49034	49.1091	7.9713
9	0.198879	9.92749	5.00555	TBA	12	38	7.35552	18.4743	49.2731	40.9865	8.28661	49.9173	8.93085
10	0.198879	10.0756	6.14201	TBA	12	38	7.59263	19.0698	49.845	40.7961	9.04893	50.6621	9.86598
11	0.198879	10.211	7.2809	TBA	12	38	7.81919	19.6388	50.3438	40.5666	9.77725	51.3429	10.7763
12	0.198879	10.3336	8.42269	TBA	12	38	8.03527	20.1816	50.7697	40.2976	10.4721	51.9595	11.6619
13	0.198879	10.4433	9.56786	TBA	12	38	8.24089	20.698	51.122	39.9889	11.1331	52.5111	12.5222
14	0.198879	10.5399	10.7169	TBA	12	38	8.43607	21.1882	51.4004	39.6399	11.7605	52.997	13.3571
15	0.198879	10.6234	11.8703	TBA	12	38	8.62081	21.6522	51.6045	39.2503	12.3542	53.4166	14.1663
16	0.198879	10.6936	13.0287	TBA	12	38	8.79508	22.0899	51.7343	38.8197	12.9146	53.7694	14.9497
17	0.198879	10.7502	14.1924	TBA	12	38	8.95888	22.5013	51.7886	38.3473	13.4413	54.0542	15.7069
18	0.198879	10.7931	15.3622	TBA	12	38	9.11217	22.8863	51.7664	37.8327	13.9337	54.2698	16.4371
19	0.198879	10.8221	16.5386	TBA	12	38	9.25478	23.2445	51.6674	37.2751	14.3923	54.4156	17.1405
20	0.198879	10.8369	17.7222	TBA	12	38	9.38677	23.576	51.4903	36.6738	14.8165	54.49	17.8162
21	0.198879	10.8372	18.9137	TBA	12	38	9.50793	23.8803	51.2341	36.0279	15.2062	54.4919	18.464
22	0.198879	10.8228	20.1138	TBA	12	38	9.61813	24.1571	50.8967	35.3365	15.5602	54.4191	19.0826
23	0.198879	10.7933	21.3231	TBA	12	38	9.71727	24.4061	50.4779	34.5986	15.8793	54.2711	19.6725
24	0.198879	10.7483	22.5424	TBA	12	38	9.80519	24.6269	49.9747	33.813	16.1617	54.0447	20.2317
25	0.198879	10.7216	23.7727	TBA	12	38	9.92873	24.9372	49.5372	32.9785	16.5587	53.9106	20.9321
26	0.198879	10.9274	25.0147	TBA	12	38	10.5637	26.532	50.0163	31.4159	18.6004	54.9455	23.5296
27	0.198879	11.166	26.2694	TBA	12	38	10.9638	27.5368	50.7337	30.8476	19.8861	56.145	25.2974
28	0.198879	11.3873	27.5378	TBA	12	38	11.3458	28.4964	51.3422	30.2276	21.1146	57.258	27.0304
29	0.198879	11.5906	28.8211	TBA	12	38	11.7095	29.4097	51.8373	29.554	22.2833	58.2802	28.7262
30	0.198879	11.7754	30.1203	TBA	12	38	12.0541	30.2753	52.2163	28.8248	23.3915	59.2095	30.3847
31	0.198879	11.9408	31.4369	TBA	12	38	12.3791	31.0915	52.4739	28.038	24.4359	60.041	32.003
32	0.198879	12.0861	32.7723	TBA	12	38	12.6837	31.8567	52.6063	27.191	25.4153	60.7717	33.5807
33	0.198879	12.2104	34.1281	TBA	12	38	12.9672	32.5688	52.6081	26.2812	26.3269	61.3969	35.1157
34	0.198879	12.3126	35.5059	TBA	12	38	13.2287	33.2254	52.4727	25.3056	27.1671	61.9107	36.6051
35	0.198879	12.3917	36.9079	TBA	12	38	13.467	33.824	52.194	24.2607	27.9333	62.3082	38.0475
36	0.198879	12.2713	38.3361	TBA	12	38	13.4613	33.8097	51.058	23.1427	27.9153	61.7029	38.5602
37	0.198879	11.768	39.7931	TBA	12	38	12.9911	32.6288	48.3509	21.9473	26.4036	59.1721	37.2248
38	0.198879	11.2266	41.2816	TBA	12	38	12.4954	31.3838	45.4796	20.6695	24.8101	56.45	35.7805
39	0.198879	10.6557	42.805	TBA	12	38	11.9866	30.1058	42.4778	19.3036	23.1742	53.5795	34.2759
40	0.198879	10.0532	44.3669	TBA	12	38	11.4638	28.7926	39.3365	17.8431	21.4934	50.5497	32.7066
41	0.198879	9.41631	45.9716	TBA	12	38	10.9259	27.4418	36.0447	16.2801	19.7646	47.3476	31.0675
42	0.198879	8.74215	47.6244	TBA	12	38	10.372	26.0504	32.5894	14.6057	17.9837	43.9578	29.3521
43	0.198879	8.02705	49.3313	TBA	12	38	9.80045	24.615	28.9555	12.8091	16.1464	40.3622	27.5531
44	0.198879	7.26663	51.0996	TBA	12	38	9.20983	23.1316	25.1249	10.877	14.2479	36.5386	25.6616
45	0.198879	6.45552	52.9386	TBA	12	38	8.59825	21.5955	21.0753	8.7936	12.2817	32.4602	23.6666
46	0.198879	5.587	54.8595	TBA	12	38	7.96335	20.0009	16.7794	6.53871	10.2407	28.0931	21.5544
47	0.198879	4.65249	56.877	TBA	12	38	7.3023	18.3406	12.2023	4.08672	8.11561	23.3942	19.3075
48	0.198879	3.6407	59.0103	TBA	12	38	6.61151	16.6056	7.2988	1.4039	5.8949	18.3067	16.9028
49	0.246868	2.96414	61.584	TBA	12	38	5.40531	13.5761	2.01729	0	2.01729	12.0075	12.0075
50	0.246868	1.03152	64.7022	TBA	12	38	3.66538	9.20604	-3.5761	0	-3.5761	4.17883	4.17883

Query 1 (bishop simplified) - Safety Factor: 2.51162

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.198879	8.2906	-4.05273	TBA	12	38	5.06716	12.7268	42.0457	41.1155	0.930247	41.6867	0.57123
2	0.198879	8.53903	-2.91954	TBA	12	38	5.3926	13.5442	43.2109	41.2344	1.9765	42.9359	1.70148
3	0.198879	8.77491	-1.7875	TBA	12	38	5.70643	14.3324	44.3001	41.3147	2.98535	44.122	2.80727
4	0.198879	8.99829	-0.656162	TBA	12	38	6.00884	15.0919	45.3139	41.3564	3.95753	45.2451	3.88871
5	0.198879	9.20916	0.474923	TBA	12	38	6.30004	15.8233	46.2532	41.3596	4.89362	46.3054	4.94584
6	0.198879	9.40754	1.60619	TBA	12	38	6.58017	16.5269	47.1184	41.3243	5.79413	47.3029	5.97865
7	0.198879	9.59341	2.73809	TBA	12	38	6.84938	17.203	47.91	41.2504	6.65959	48.2376	6.98716
8	0.198879	9.76674	3.87106	TBA	12	38	7.10779	17.8521	48.6281	41.1378	7.49034	49.1091	7.9713
9	0.198879	9.92749	5.00555	TBA	12	38	7.35552	18.4743	49.2731	40.9865	8.28661	49.9173	8.93085
10	0.198879	10.0756	6.14201	TBA	12	38	7.59263	19.0698	49.845	40.7961	9.04893	50.6621	9.86598
11	0.198879	10.211	7.2809	TBA	12	38	7.81919	19.6388	50.3438	40.5666	9.77725	51.3429	10.7763
12	0.198879	10.3336	8.42269	TBA	12	38	8.03527	20.1816	50.7697	40.2976	10.4721	51.9595	11.6619
13	0.198879	10.4433	9.56786	TBA	12	38	8.24089	20.698	51.122	39.9889	11.1331	52.5111	12.5222
14	0.198879	10.5399	10.7169	TBA	12	38	8.43607	21.1882	51.4004	39.6399	11.7605	52.997	13.3571
15	0.198879	10.6234	11.8703	TBA	12	38	8.62081	21.6522	51.6045	39.2503	12.3542	53.4166	14.1663
16	0.198879	10.6936	13.0287	TBA	12	38	8.79508	22.0899	51.7343	38.8197	12.9146	53.7694	14.9497
17	0.198879	10.7502	14.1924	TBA	12	38	8.95888	22.5013	51.7886	38.3473	13.4413	54.0542	15.7069
18	0.198879	10.7931	15.3622	TBA	12	38	9.11217	22.8863	51.7664	37.8327	13.9337	54.2698	16.4371
19	0.198879	10.8221	16.5386	TBA	12	38	9.25478	23.2445	51.6674	37.2751	14.3923	54.4156	17.1405
20	0.198879	10.8369	17.7222	TBA	12	38	9.38677	23.576	51.4903	36.6738	14.8165	54.49	17.8162
21	0.198879	10.8372	18.9137	TBA	12	38	9.50793	23.8803	51.2341	36.0279	15.2062	54.4919	18.464
22	0.198879	10.8228	20.1138	TBA	12	38	9.61813	24.1571	50.8967	35.3365	15.5602	54.4191	19.0826
23	0.198879	10.7933	21.3231	TBA	12	38	9.71727	24.4061	50.4779	34.5986	15.8793	54.2711	19.6725
24	0.198879	10.7483	22.5424	TBA	12	38	9.80519	24.6269	49.9747	33.813	16.1617	54.0447	20.2317
25	0.198879	10.7216	23.7727	TBA	12	38	9.92873	24.9372	49.5372	32.9785	16.5587	53.9106	20.9321
26	0.198879	10.9274	25.0147	TBA	12	38	10.5637	26.532	50.0163	31.4159	18.6004	54.9455	23.5296
27	0.198879	11.166	26.2694	TBA	12	38	10.9638	27.5368	50.7337	30.8476	19.8861	56.145	25.2974
28	0.198879	11.3873	27.5378	TBA	12	38	11.3458	28.4964	51.3422	30.2276	21.1146	57.258	27.0304
29	0.198879	11.5906	28.8211	TBA	12	38	11.7095	29.4097	51.8373	29.554	22.2833	58.2802	28.7262
30	0.198879	11.7754	30.1203	TBA	12	38	12.0541	30.2753	52.2163	28.8248	23.3915	59.2095	30.3847
31	0.198879	11.9408	31.4369	TBA	12	38	12.3791	31.0915	52.4739	28.038	24.4359	60.041	32.003
32	0.198879	12.0861	32.7723	TBA	12	38	12.6837	31.8567	52.6063	27.191	25.4153	60.7717	33.5807
33	0.198879	12.2104	34.1281	TBA	12	38	12.9672	32.5688	52.6081	26.2812	26.3269	61.3969	35.1157
34	0.198879	12.3126	35.5059	TBA	12	38	13.2287	33.2254	52.4727	25.3056	27.1671	61.9107	36.6051
35	0.198879	12.3917	36.9079	TBA	12	38	13.467	33.824	52.194	24.2607	27.9333	62.3082	38.0475
36	0.198879	12.2713	38.3361	TBA	12	38	13.4613	33.8097	51.058	23.1427	27.9153	61.7029	38.5602
37	0.198879	11.768	39.7931	TBA	12	38	12.9911	32.6288	48.3509	21.9473	26.4036	59.1721	37.2248
38	0.198879	11.2266	41.2816	TBA	12	38	12.4954	31.3838	45.4796	20.6695	24.8101	56.45	35.7805
39	0.198879	10.6557	42.805	TBA	12	38	11.9866	30.1058	42.4778	19.3036	23.1742	53.5795	34.2759
40	0.198879	10.0532	44.3669	TBA	12	38	11.4638	28.7926	39.3365	17.8431	21.4934	50.5497	32.7066
41	0.198879	9.41631	45.9716	TBA	12	38	10.9259	27.4418	36.0447	16.2801	19.7646	47.3476	31.0675
42	0.198879	8.74215	47.6244	TBA	12	38	10.372	26.0504	32.5894	14.6057	17.9837	43.9578	29.3521
43	0.198879	8.02705	49.3313	TBA	12	38	9.80045	24.615	28.9555	12.8091	16.1464	40.3622	27.5531
44	0.198879	7.26663	51.0996	TBA	12	38	9.20983	23.1316	25.1249	10.877	14.2479	36.5386	25.6616
45	0.198879	6.45552	52.9386	TBA	12	38	8.59825	21.5955	21.0753	8.7936	12.2817	32.4602	23.6666
46	0.198879	5.587	54.8595	TBA	12	38	7.96335	20.0009	16.7794	6.53871	10.2407	28.0931	21.5544
47	0.198879	4.65249	56.877	TBA	12	38	7.3023	18.3406	12.2023	4.08672	8.11561	23.3942	19.3075
48	0.198879	3.6407	59.0103	TBA	12	38	6.61151	16.6056	7.2988	1.4039	5.8949	18.3067	16.9028
49	0.246868	2.96414	61.584	TBA	12	38	5.40531	13.5761	2.01729	0	2.01729	12.0075	12.0075
50	0.246868	1.03152	64.7022	TBA	12	38	3.66538	9.20604	-3.5761	0	-3.5761	4.17883	4.17883

Global Minimum Query (spencer) - Safety Factor: 2.50591

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.198879	8.2906	-4.05273	TBA	12	38	7.52238	18.8504	49.8836	41.1155	8.76811	49.3506	8.23513
2	0.198879	8.53903	-2.91954	TBA	12	38	7.71633	19.3364	50.6244	41.2344	9.39003	50.2309	8.9965
3	0.198879	8.77491	-1.7875	TBA	12	38	7.89812	19.792	51.2883	41.3147	9.97355	51.0418	9.72707
4	0.198879	8.99829	-0.656162	TBA	12	38	8.06817	20.2181	51.875	41.3564	10.5186	51.7826	10.4262
5	0.198879	9.20916	0.474923	TBA	12	38	8.22686	20.6158	52.3875	41.3596	11.0279	52.4556	11.096
6	0.198879	9.40754	1.60619	TBA	12	38	8.3745	20.9857	52.8256	41.3243	11.5013	53.0604	11.7361
7	0.198879	9.59341	2.73809	TBA	12	38	8.51142	21.3289	53.1908	41.2504	11.9404	53.5979	12.3475
8	0.198879	9.76674	3.87106	TBA	12	38	8.6379	21.6458	53.4838	41.1378	12.346	54.0683	12.9305
9	0.198879	9.92749	5.00555	TBA	12	38	8.75419	21.9372	53.7057	40.9865	12.7192	54.4724	13.4859
10	0.198879	10.0756	6.14201	TBA	12	38	8.86053	22.2037	53.8563	40.7961	13.0602	54.8098	14.0137
11	0.198879	10.211	7.2809	TBA	12	38	8.95715	22.4458	53.9365	40.5666	13.3699	55.0809	14.5143
12	0.198879	10.3336	8.42269	TBA	12	38	9.04418	22.6639	53.9471	40.2976	13.6495	55.2863	14.9887
13	0.198879	10.4433	9.56786	TBA	12	38	9.12188	22.8586	53.8872	39.9889	13.8983	55.4248	15.4359
14	0.198879	10.5399	10.7169	TBA	12	38	9.19039	23.0303	53.758	39.6399	14.1181	55.4973	15.8574
15	0.198879	10.6234	11.8703	TBA	12	38	9.24981	23.1792	53.559	39.2503	14.3087	55.5033	16.253
16	0.198879	10.6936	13.0287	TBA	12	38	9.30029	23.3057	53.2901	38.8197	14.4704	55.4422	16.6225
17	0.198879	10.7502	14.1924	TBA	12	38	9.342	23.4102	52.9517	38.3473	14.6044	55.3142	16.9669
18	0.198879	10.7931	15.3622	TBA	12	38	9.37496	23.4928	52.5426	37.8327	14.7099	55.1183	17.2856
19	0.198879	10.8221	16.5386	TBA	12	38	9.3993	23.5538	52.0631	37.2751	14.788	54.8542	17.5791
20	0.198879	10.8369	17.7222	TBA	12	38	9.4151	23.5934	51.5128	36.6738	14.839	54.5216	17.8478
21	0.198879	10.8372	18.9137	TBA	12	38	9.42241	23.6117	50.8901	36.0279	14.8622	54.1187	18.0908
22	0.198879	10.8228	20.1138	TBA	12	38	9.42129	23.6089	50.1951	35.3365	14.8586	53.6454	18.3089
23	0.198879	10.7933	21.3231	TBA	12	38	9.41179	23.5851	49.4268	34.5986	14.8282	53.1007	18.5021
24	0.198879	10.7483	22.5424	TBA	12	38	9.39395	23.5404	48.5839	33.813	14.7709	52.4831	18.6701
25	0.198879	10.7216	23.7727	TBA	12	38	9.42069	23.6074	47.8351	32.9785	14.8566	51.9847	19.0062
26	0.198879	10.9274	25.0147	TBA	12	38	10.0134	25.0926	48.1736	31.4159	16.7577	52.8461	21.4302
27	0.198879	11.166	26.2694	TBA	12	38	10.3627	25.9679	48.7258	30.8476	17.8782	53.8405	22.9929
28	0.198879	11.3873	27.5378	TBA	12	38	10.6933	26.7964	49.166	30.2276	18.9384	54.7415	24.5139
29	0.198879	11.5906	28.8211	TBA	12	38	11.005	27.5776	49.4923	29.554	19.9383	55.5477	25.9937
30	0.198879	11.7754	30.1203	TBA	12	38	11.2977	28.3109	49.702	28.8248	20.8772	56.2563	27.4315
31	0.198879	11.9408	31.4369	TBA	12	38	11.5708	28.9955	49.7912	28.038	21.7532	56.8644	28.8264
32	0.198879	12.0861	32.7723	TBA	12	38	11.8242	29.6304	49.7567	27.191	22.5657	57.3688	30.1778
33	0.198879	12.2104	34.1281	TBA	12	38	12.0573	30.2145	49.5947	26.2812	23.3135	57.7667	31.4855
34	0.198879	12.3126	35.5059	TBA	12	38	12.2696	30.7466	49.3001	25.3056	23.9945	58.0538	32.7482
35	0.198879	12.3917	36.9079	TBA	12	38	12.4606	31.2251	48.8678	24.2607	24.6071	58.2262	33.9655
36	0.198879	12.2713	38.3361	TBA	12	38	12.4228	31.1303	47.6283	23.1427	24.4856	57.4519	34.3092
37	0.198879	11.768	39.7931	TBA	12	38	11.953	29.9532	44.9263	21.9473	22.979	54.8828	32.9355
38	0.198879	11.2266	41.2816	TBA	12	38	11.4671	28.7355	42.0902	20.6695	21.4207	52.1578	31.4883
39	0.198879	10.6557	42.805	TBA	12	38	10.9775	27.5086	39.1536	19.3036	19.85	49.3207	30.0171
40	0.198879	10.0532	44.3669	TBA	12	38	10.4842	26.2725	36.1111	17.8431	18.268	46.3661	28.523
41	0.198879	9.41631	45.9716	TBA	12	38	9.98751	25.0278	32.9549	16.2801	16.6748	43.287	27.0069
42	0.198879	8.74215	47.6244	TBA	12	38	9.48753	23.7749	29.677	14.6057	15.0713	40.0761	25.4704
43	0.198879	8.02705	49.3313	TBA	12	38	8.98472	22.5149	26.2675	12.8091	13.4584	36.7248	23.9157
44	0.198879	7.26663	51.0996	TBA	12	38	8.47953	21.2489	22.7151	10.877	11.8381	33.2238	22.3468
45	0.198879	6.45552	52.9386	TBA	12	38	7.97271	19.9789	19.0061	8.7936	10.2125	29.5627	20.7691
46	0.198879	5.587	54.8595	TBA	12	38	7.46537	18.7076	15.124	6.53871	8.58528	25.7302	19.1915
47	0.198879	4.65249	56.877	TBA	12	38	6.95913	17.4389	11.0483	4.08672	6.96154	21.7142	17.6275
48	0.198879	3.6407	59.0103	TBA	12	38	6.45646	16.1793	6.75316	1.4039	5.34926	17.5029	16.099
49	0.246868	2.96414	61.584	TBA	12	38	5.46335	13.6907	2.16394	0	2.16394	12.2614	12.2614
50	0.246868	1.03152	64.7022	TBA	12	38	3.97457	9.95991	-2.6112	0	-2.6112	5.79788	5.79788

Query 1 (spencer) - Safety Factor: 2.50591

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.198879	8.2906	-4.05273	TBA	12	38	7.52238	18.8504	49.8836	41.1155	8.76811	49.3506	8.23513
2	0.198879	8.53903	-2.91954	TBA	12	38	7.71633	19.3364	50.6244	41.2344	9.39003	50.2309	8.9965
3	0.198879	8.77491	-1.7875	TBA	12	38	7.89812	19.792	51.2883	41.3147	9.97355	51.0418	9.72707
4	0.198879	8.99829	-0.656162	TBA	12	38	8.06817	20.2181	51.875	41.3564	10.5186	51.7826	10.4262
5	0.198879	9.20916	0.474923	TBA	12	38	8.22686	20.6158	52.3875	41.3596	11.0279	52.4556	11.096
6	0.198879	9.40754	1.60619	TBA	12	38	8.3745	20.9857	52.8256	41.3243	11.5013	53.0604	11.7361
7	0.198879	9.59341	2.73809	TBA	12	38	8.51142	21.3289	53.1908	41.2504	11.9404	53.5979	12.3475
8	0.198879	9.76674	3.87106	TBA	12	38	8.6379	21.6458	53.4838	41.1378	12.346	54.0683	12.9305
9	0.198879	9.92749	5.00555	TBA	12	38	8.75419	21.9372	53.7057	40.9865	12.7192	54.4724	13.4859
10	0.198879	10.0756	6.14201	TBA	12	38	8.86053	22.2037	53.8563	40.7961	13.0602	54.8098	14.0137
11	0.198879	10.211	7.2809	TBA	12	38	8.95715	22.4458	53.9365	40.5666	13.3699	55.0809	14.5143
12	0.198879	10.3336	8.42269	TBA	12	38	9.04418	22.6639	53.9471	40.2976	13.6495	55.2863	14.9887
13	0.198879	10.4433	9.56786	TBA	12	38	9.12188	22.8586	53.8872	39.9889	13.8983	55.4248	15.4359
14	0.198879	10.5399	10.7169	TBA	12	38	9.19039	23.0303	53.758	39.6399	14.1181	55.4973	15.8574
15	0.198879	10.6234	11.8703	TBA	12	38	9.24981	23.1792	53.559	39.2503	14.3087	55.5033	16.253
16	0.198879	10.6936	13.0287	TBA	12	38	9.30029	23.3057	53.2901	38.8197	14.4704	55.4422	16.6225
17	0.198879	10.7502	14.1924	TBA	12	38	9.342	23.4102	52.9517	38.3473	14.6044	55.3142	16.9669
18	0.198879	10.7931	15.3622	TBA	12	38	9.37496	23.4928	52.5426	37.8327	14.7099	55.1183	17.2856
19	0.198879	10.8221	16.5386	TBA	12	38	9.3993	23.5538	52.0631	37.2751	14.788	54.8542	17.5791
20	0.198879	10.8369	17.7222	TBA	12	38	9.4151	23.5934	51.5128	36.6738	14.839	54.5216	17.8478
21	0.198879	10.8372	18.9137	TBA	12	38	9.42241	23.6117	50.8901	36.0279	14.8622	54.1187	18.0908
22	0.198879	10.8228	20.1138	TBA	12	38	9.42129	23.6089	50.1951	35.3365	14.8586	53.6454	18.3089
23	0.198879	10.7933	21.3231	TBA	12	38	9.41179	23.5851	49.4268	34.5986	14.8282	53.1007	18.5021
24	0.198879	10.7483	22.5424	TBA	12	38	9.39395	23.5404	48.5839	33.813	14.7709	52.4831	18.6701
25	0.198879	10.7216	23.7727	TBA	12	38	9.42069	23.6074	47.8351	32.9785	14.8566	51.9847	19.0062
26	0.198879	10.9274	25.0147	TBA	12	38	10.0134	25.0926	48.1736	31.4159	16.7577	52.8461	21.4302
27	0.198879	11.166	26.2694	TBA	12	38	10.3627	25.9679	48.7258	30.8476	17.8782	53.8405	22.9929
28	0.198879	11.3873	27.5378	TBA	12	38	10.6933	26.7964	49.166	30.2276	18.9384	54.7415	24.5139
29	0.198879	11.5906	28.8211	TBA	12	38	11.005	27.5776	49.4923	29.554	19.9383	55.5477	25.9937
30	0.198879	11.7754	30.1203	TBA	12	38	11.2977	28.3109	49.702	28.8248	20.8772	56.2563	27.4315
31	0.198879	11.9408	31.4369	TBA	12	38	11.5708	28.9955	49.7912	28.038	21.7532	56.8644	28.8264
32	0.198879	12.0861	32.7723	TBA	12	38	11.8242	29.6304	49.7567	27.191	22.5657	57.3688	30.1778
33	0.198879	12.2104	34.1281	TBA	12	38	12.0573	30.2145	49.5947	26.2812	23.3135	57.7667	31.4855
34	0.198879	12.3126	35.5059	TBA	12	38	12.2696	30.7466	49.3001	25.3056	23.9945	58.0538	32.7482
35	0.198879	12.3917	36.9079	TBA	12	38	12.4606	31.2251	48.8678	24.2607	24.6071	58.2262	33.9655
36	0.198879	12.2713	38.3361	TBA	12	38	12.4228	31.1303	47.6283	23.1427	24.4856	57.4519	34.3092
37	0.198879	11.768	39.7931	TBA	12	38	11.953	29.9532	44.9263	21.9473	22.979	54.8828	32.9355
38	0.198879	11.2266	41.2816	TBA	12	38	11.4671	28.7355	42.0902	20.6695	21.4207	52.1578	31.4883
39	0.198879	10.6557	42.805	TBA	12	38	10.9775	27.5086	39.1536	19.3036	19.85	49.3207	30.0171
40	0.198879	10.0532	44.3669	TBA	12	38	10.4842	26.2725	36.1111	17.8431	18.268	46.3661	28.523
41	0.198879	9.41631	45.9716	TBA	12	38	9.98751	25.0278	32.9549	16.2801	16.6748	43.287	27.0069
42	0.198879	8.74215	47.6244	TBA	12	38	9.48753	23.7749	29.677	14.6057	15.0713	40.0761	25.4704
43	0.198879	8.02705	49.3313	TBA	12	38	8.98472	22.5149	26.2675	12.8091	13.4584	36.7248	23.9157
44	0.198879	7.26663	51.0996	TBA	12	38	8.47953	21.2489	22.7151	10.877	11.8381	33.2238	22.3468
45	0.198879	6.45552	52.9386	TBA	12	38	7.97271	19.9789	19.0061	8.7936	10.2125	29.5627	20.7691
46	0.198879	5.587	54.8595	TBA	12	38	7.46537	18.7076	15.124	6.53871	8.58528	25.7302	19.1915
47	0.198879	4.65249	56.877	TBA	12	38	6.95913	17.4389	11.0483	4.08672	6.96154	21.7142	17.6275
48	0.198879	3.6407	59.0103	TBA	12	38	6.45646	16.1793	6.75316	1.4039	5.34926	17.5029	16.099
49	0.246868	2.96414	61.584	TBA	12	38	5.46335	13.6907	2.16394	0	2.16394	12.2614	12.2614
50	0.246868	1.03152	64.7022	TBA	12	38	3.97457	9.95991	-2.6112	0	-2.6112	5.79788	5.79788

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 2.51162

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	5.00013	3.00011	85.8717	0	0
2	5.19901	2.98602	8.45431	0	0
3	5.39789	2.97588	16.5341	0	0
4	5.59676	2.96967	24.228	0	0
5	5.79564	2.96739	31.5252	0	0
6	5.99452	2.96904	38.4158	0	0
7	6.1934	2.97462	44.8906	0	0
8	6.39228	2.98413	50.9409	0	0
9	6.59116	2.99759	56.5589	0	0
10	6.79004	3.015	61.7373	0	0
11	6.98891	3.03641	66.4693	0	0
12	7.18779	3.06182	70.7488	0	0
13	7.38667	3.09126	74.5704	0	0
14	7.58555	3.12479	77.9292	0	0
15	7.78443	3.16243	80.8209	0	0
16	7.98331	3.20423	83.2417	0	0
17	8.18219	3.25025	85.1886	0	0
18	8.38106	3.30055	86.659	0	0
19	8.57994	3.35518	87.6511	0	0
20	8.77882	3.41424	88.1638	0	0
21	8.9777	3.4778	88.1965	0	0
22	9.17658	3.54594	87.7494	0	0
23	9.37546	3.61877	86.8235	0	0
24	9.57434	3.69641	85.4206	0	0
25	9.77321	3.77896	83.5434	0	0
26	9.97209	3.86656	81.2209	0	0
27	10.171	3.95936	78.6801	0	0
28	10.3698	4.05752	75.8804	0	0
29	10.5687	4.16122	72.8128	0	0
30	10.7676	4.27065	69.4689	0	0
31	10.9665	4.38603	65.8414	0	0
32	11.1654	4.5076	61.9238	0	0
33	11.3642	4.63563	57.7109	0	0
34	11.5631	4.77042	53.1984	0	0
35	11.762	4.91231	48.3838	0	0
36	11.9609	5.06168	43.266	0	0
37	12.1598	5.21895	37.9133	0	0
38	12.3586	5.38461	32.4871	0	0
39	12.5575	5.55921	27.031	0	0
40	12.7564	5.74341	21.5905	0	0
41	12.9553	5.93794	16.2181	0	0
42	13.1541	6.14368	10.975	0	0
43	13.353	6.36167	5.93362	0	0
44	13.5519	6.59314	1.1802	0	0
45	13.7508	6.83961	-3.18076	0	0
46	13.9497	7.10294	-7.02067	0	0
47	14.1485	7.38549	-10.178	0	0
48	14.3474	7.6903	-12.4453	0	0
49	14.5463	8.02143	-13.5472	0	0
50	14.7932	8.4777	-13.1333	0	0
51	15.04	9	0	0	0

Query 1 (bishop simplified) - Safety Factor: 2.51162

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	5.00013	3.00011	85.8717	0	0
2	5.19901	2.98602	8.45431	0	0
3	5.39789	2.97588	16.5341	0	0
4	5.59676	2.96967	24.228	0	0
5	5.79564	2.96739	31.5252	0	0
6	5.99452	2.96904	38.4158	0	0
7	6.1934	2.97462	44.8906	0	0
8	6.39228	2.98413	50.9409	0	0
9	6.59116	2.99759	56.5589	0	0
10	6.79004	3.015	61.7373	0	0
11	6.98891	3.03641	66.4693	0	0
12	7.18779	3.06182	70.7488	0	0
13	7.38667	3.09126	74.5704	0	0
14	7.58555	3.12479	77.9292	0	0
15	7.78443	3.16243	80.8209	0	0
16	7.98331	3.20423	83.2417	0	0
17	8.18219	3.25025	85.1886	0	0
18	8.38106	3.30055	86.659	0	0
19	8.57994	3.35518	87.6511	0	0
20	8.77882	3.41424	88.1638	0	0
21	8.9777	3.4778	88.1965	0	0
22	9.17658	3.54594	87.7494	0	0
23	9.37546	3.61877	86.8235	0	0
24	9.57434	3.69641	85.4206	0	0
25	9.77321	3.77896	83.5434	0	0
26	9.97209	3.86656	81.2209	0	0
27	10.171	3.95936	78.6801	0	0
28	10.3698	4.05752	75.8804	0	0
29	10.5687	4.16122	72.8128	0	0
30	10.7676	4.27065	69.4689	0	0
31	10.9665	4.38603	65.8414	0	0
32	11.1654	4.5076	61.9238	0	0
33	11.3642	4.63563	57.7109	0	0
34	11.5631	4.77042	53.1984	0	0
35	11.762	4.91231	48.3838	0	0
36	11.9609	5.06168	43.266	0	0
37	12.1598	5.21895	37.9133	0	0
38	12.3586	5.38461	32.4871	0	0
39	12.5575	5.55921	27.031	0	0
40	12.7564	5.74341	21.5905	0	0
41	12.9553	5.93794	16.2181	0	0
42	13.1541	6.14368	10.975	0	0
43	13.353	6.36167	5.93362	0	0
44	13.5519	6.59314	1.1802	0	0
45	13.7508	6.83961	-3.18076	0	0
46	13.9497	7.10294	-7.02067	0	0
47	14.1485	7.38549	-10.178	0	0
48	14.3474	7.6903	-12.4453	0	0
49	14.5463	8.02143	-13.5472	0	0
50	14.7932	8.4777	-13.1333	0	0
51	15.04	9	0	0	0

Global Minimum Query (spencer) - Safety Factor: 2.50591

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	5.00013	3.00011	85.8717	0	0
2	5.19901	2.98602	9.05292	1.52451	9.55891
3	5.39789	2.97588	17.6699	2.97562	9.55895
4	5.59676	2.96967	25.8429	4.35195	9.55894
5	5.79564	2.96739	33.5645	5.65226	9.55892
6	5.99452	2.96904	40.8281	6.87545	9.55892
7	6.1934	2.97462	47.6278	8.02052	9.55892
8	6.39228	2.98413	53.9584	9.08659	9.55892
9	6.59116	2.99759	59.8152	10.0729	9.55894
10	6.79004	3.015	65.1944	10.9787	9.55889
11	6.98891	3.03641	70.0926	11.8036	9.55893
12	7.18779	3.06182	74.507	12.547	9.55894
13	7.38667	3.09126	78.4356	13.2085	9.55888
14	7.58555	3.12479	81.8768	13.788	9.55889
15	7.78443	3.16243	84.8296	14.2853	9.55892
16	7.98331	3.20423	87.2937	14.7002	9.55889
17	8.18219	3.25025	89.2693	15.0329	9.55889
18	8.38106	3.30055	90.7573	15.2835	9.55891
19	8.57994	3.35518	91.7592	15.4522	9.55889
20	8.77882	3.41424	92.2771	15.5395	9.55895
21	8.9777	3.4778	92.3138	15.5456	9.5589
22	9.17658	3.54594	91.8731	15.4714	9.55891
23	9.37546	3.61877	90.959	15.3175	9.55893
24	9.57434	3.69641	89.5768	15.0847	9.5589
25	9.77321	3.77896	87.7325	14.7741	9.55889
26	9.97209	3.86656	85.4579	14.3911	9.55892
27	10.171	3.95936	82.9786	13.9736	9.55893
28	10.3698	4.05752	80.2564	13.5152	9.55894
29	10.5687	4.16122	77.2845	13.0147	9.55892
30	10.7676	4.27065	74.0569	12.4712	9.55894
31	10.9665	4.38603	70.5689	11.8838	9.55892
32	11.1654	4.5076	66.8167	11.2519	9.5589
33	11.3642	4.63563	62.7975	10.5751	9.55892
34	11.5631	4.77042	58.5101	9.8531	9.55892
35	11.762	4.91231	53.9548	9.08599	9.55893
36	11.9609	5.06168	49.1336	8.27409	9.55891
37	12.1598	5.21895	44.1135	7.42871	9.55892
38	12.3586	5.38461	39.048	6.57568	9.55892
39	12.5575	5.55921	33.9791	5.72208	9.55892
40	12.7564	5.74341	28.9501	4.8752	9.55893
41	12.9553	5.93794	24.0102	4.04332	9.55893
42	13.1541	6.14368	19.2161	3.236	9.55895
43	13.353	6.36167	14.6336	2.4643	9.55893
44	13.5519	6.59314	10.3401	1.74127	9.55891
45	13.7508	6.83961	6.42768	1.08242	9.55891
46	13.9497	7.10294	3.00816	0.506574	9.55892
47	14.1485	7.38549	0.219405	0.0369479	9.55895
48	14.3474	7.6903	-1.76436	-0.297118	9.55892
49	14.5463	8.02143	-2.71659	-0.457474	9.55893
50	14.7932	8.4777	-2.35535	-0.396641	9.55892
51	15.04	9	0	0	0

Query 1 (spencer) - Safety Factor: 2.50591

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	5.00013	3.00011	85.8717	0	0
2	5.19901	2.98602	9.05292	1.52451	9.55891
3	5.39789	2.97588	17.6699	2.97562	9.55895
4	5.59676	2.96967	25.8429	4.35195	9.55894
5	5.79564	2.96739	33.5645	5.65226	9.55892
6	5.99452	2.96904	40.8281	6.87545	9.55892
7	6.1934	2.97462	47.6278	8.02052	9.55892
8	6.39228	2.98413	53.9584	9.08659	9.55892
9	6.59116	2.99759	59.8152	10.0729	9.55894
10	6.79004	3.015	65.1944	10.9787	9.55889
11	6.98891	3.03641	70.0926	11.8036	9.55893
12	7.18779	3.06182	74.507	12.547	9.55894
13	7.38667	3.09126	78.4356	13.2085	9.55888
14	7.58555	3.12479	81.8768	13.788	9.55889
15	7.78443	3.16243	84.8296	14.2853	9.55892
16	7.98331	3.20423	87.2937	14.7002	9.55889
17	8.18219	3.25025	89.2693	15.0329	9.55889
18	8.38106	3.30055	90.7573	15.2835	9.55891
19	8.57994	3.35518	91.7592	15.4522	9.55889
20	8.77882	3.41424	92.2771	15.5395	9.55895
21	8.9777	3.4778	92.3138	15.5456	9.5589
22	9.17658	3.54594	91.8731	15.4714	9.55891
23	9.37546	3.61877	90.959	15.3175	9.55893
24	9.57434	3.69641	89.5768	15.0847	9.5589
25	9.77321	3.77896	87.7325	14.7741	9.55889
26	9.97209	3.86656	85.4579	14.3911	9.55892
27	10.171	3.95936	82.9786	13.9736	9.55893
28	10.3698	4.05752	80.2564	13.5152	9.55894
29	10.5687	4.16122	77.2845	13.0147	9.55892
30	10.7676	4.27065	74.0569	12.4712	9.55894
31	10.9665	4.38603	70.5689	11.8838	9.55892
32	11.1654	4.5076	66.8167	11.2519	9.5589
33	11.3642	4.63563	62.7975	10.5751	9.55892
34	11.5631	4.77042	58.5101	9.8531	9.55892
35	11.762	4.91231	53.9548	9.08599	9.55893
36	11.9609	5.06168	49.1336	8.27409	9.55891
37	12.1598	5.21895	44.1135	7.42871	9.55892
38	12.3586	5.38461	39.048	6.57568	9.55892
39	12.5575	5.55921	33.9791	5.72208	9.55892
40	12.7564	5.74341	28.9501	4.8752	9.55893
41	12.9553	5.93794	24.0102	4.04332	9.55893
42	13.1541	6.14368	19.2161	3.236	9.55895
43	13.353	6.36167	14.6336	2.4643	9.55893
44	13.5519	6.59314	10.3401	1.74127	9.55891
45	13.7508	6.83961	6.42768	1.08242	9.55891
46	13.9497	7.10294	3.00816	0.506574	9.55892
47	14.1485	7.38549	0.219405	0.0369479	9.55895
48	14.3474	7.6903	-1.76436	-0.297118	9.55892
49	14.5463	8.02143	-2.71659	-0.457474	9.55893
50	14.7932	8.4777	-2.35535	-0.396641	9.55892
51	15.04	9	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	0, 0 20, 0 20, 9 12, 9 5, 3 0, 3

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Water Table	0, 7.184 9.8819, 7.18448 20, 9	Assigned to: TBA